EAST WEST UNIVERSITY

Lab 4

Incremental Pattern Mining & Correlation on YouTube Comments

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Introduction

This week's lab work, "*Incremental Pattern Mining and Correlation on YouTube Comments*", builds on the earlier data cleaning and preprocessing work completed in previous labs. The main objective is to explore how frequent textual patterns evolve as new data becomes available over time. The cleaned YouTube comments created from Lab 1 "*Exploratory Data Analysis on YouTube Data"* represent the incoming flow of online user activity.

The experiment examines the most common words and word pairs through separate analysis of each segment while showing their gradual increase in frequency. The method demonstrates a practical data mining procedure which needs to perform ongoing updates instead of complete recalculations. The application of correlation analysis enables researchers to identify connections between repeated words and themes which shows how different concepts develop together in the dataset.

This report demonstrates how static text data becomes a time-based analytical system which helps researchers understand comment changes and pattern development in YouTube comment sections.

Objectives

- To create simulation of YouTube comment arrival in increments.
- To identify and visualize the top 10 most frequent individual words (unigrams) and word pairs (co-occurring itemsets) within each chunk.
- To track and analysis pattern development by following frequency changes throughout the data chunks.
- To calculate pattern correlations to identify relationships or dependencies that exist between them.
- To interpret the discovered patterns and relationships to understand how discussion topics and user sentiments have evolved throughout the period.

Tools and Libraries

- **Python 3.10** for running all the code.

- **pandas** – for creating and managing DataFrames, reading the CSVs, and organizing word frequency data.
- **numpy** – for splitting the dataset into equal segments using array_split() to simulate incremental data flow.
- **collections.Counter** – to efficiently track frequency unigrams and co-occurring word pairs.
- **itertools.combinations** – to generate word pairs within each comment.
- **matplotlib.pyplot** – for visualizing the frequency distributions of unigrams and bigrams using bar charts and line graphs.
- **nltk** – for linguistic bigram generation and token management.
- The whole setup was done on Google Colab.

Dataset Description

The dataset used in this lab, cleaned_comments.csv, was the cleaned output from the earlier data-preprocessing stage (Lab 2). The dataset includes normalized and tokenized YouTube comments which appear in the `cleaned_tokens` column as lists of words that represent individual comment segments.

The dataset was split into five equal parts which created a situation that duplicated how new comment data appears in real-world applications. The data divides into separate sections which show how information flows in chronological order. The `cleaned_tokens` column served as the basis for generating transactions in which frequent words (unigrams) and co-occurring pairs were identified using pattern-mining techniques.

The lab report explores to study single word usage patterns and how words link together throughout different time intervals. The frequency monitors how common expressions and themes changed over time while their correlation analysis revealed which recurring terms showed simultaneous changes thus enabling them to understand comment dynamics and user behavior patterns more deeply.

Methodology

Data Loading and Preparation

The **pandas** library imported the **cleaned_comments.csv** file which contains **963** YouTube comments. Each row represented a unique comment,and the column `cleaned_tokens` stored a preprocessed list of words for that comment. The tokens existed in string form as list representations which required conversion through `ast.literal_eval()` to transform them into Python lists. A sanity check confirmed that the data was properly formatted, displaying both the row count and a sample of tokenized words. The dataset maintained extra columns like username, timestamp and likes although remained unused for pattern analysis. The analysis focused exclusively on textual content from the `cleaned_tokens` field.

Data Segmentation

The dataset is divided into five equal parts using NumPy's `array_split()` function to simulate the step-by-step arrival of new comment data.

Chunk 1	Chunk 2	Chunk 3	Chunk 4	Chunk 5
44	43	43	43	43

Table 1. Number of Comments per Chunk

The analysis of each section became possible through this division which showed the development of word frequencies and word pair patterns when new data sets were added.

Pattern Mining and Visualization

Each chunk's `cleaned_tokens` were treated as **transactions** of words. Two types of frequent patterns were extracted:

- **Unigrams (single word):** The Python `collections.Counter` class enables users to count single words which function as unigrams.
- **Co-occurring pairs (unordered pairs):** The unordered pairs of co-occurring terms emerged through `itertools.combinations()` which identifies all unique token combinations within each comment

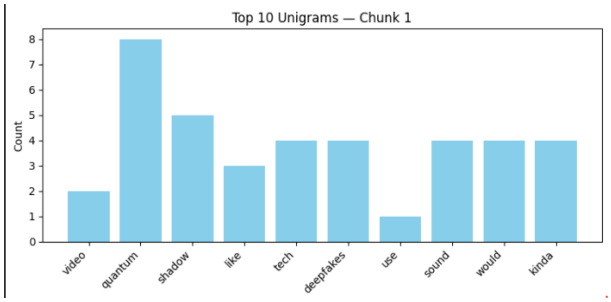

Fig. 1 Top 10 Unigrams for Chunk 1

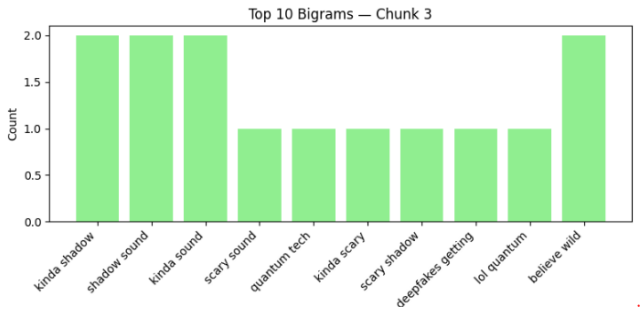

Fig. 2 Top 10 Co-occurring Word Pairs for Chunks 3

Frequency Tracking Tables

Two frequency-tracking tables were generated to summarize the evolution of the most common terms across all chunks. Missing words in a chunk were assigned to a frequency of zero.

Top 10 Unigrams	**Chunk 1**	**Chunk 2**	**Chunk 3**	**Chunk 4**	**Chunk 5**
Video	2	3	6	10	10
Quantum	8	9	5	3	1
Shadow	5	9	3	1	0
like	3	6	7	2	0
tech	4	7	4	0	1
deepfakes	4	10	1	0	0
use	1	0	7	5	3
sound	4	8	3	0	1
would	4	1	4	3	3
kinda	4	9	2	0	0

Table 2. Unigram Frequency Tracking Table

Similarly, co-occurring pair frequencies were tracked across chunks:

Pattern	Chunk 1	Chunk 2	Chunk 3	Chunk 4	Chunk 5
(kinda, shadow)	4	6	2	0	0
(shadow, sound)	3	6	2	0	0
(kinda, sound)	3	5	2	0	0
(scary, sound)	2	5	1	0	0
(quantum, tech)	2	4	1	0	0
(kinda, scary)	2	4	1	0	0
(scary, shadow)	2	4	1	0	0
(deepfakes, getting)	1	5	1	0	0
(lol, quantum)	2	3	1	0	0
(believe, wild)	2	2	2	0	0

Table 3: Co-occurring Pair Frequency Tracking Table

Correlation Analysis

The analysis of trends and relationships among selected tokens required choosing 10 unigrams and 10 bigrams. Frequency values for each pattern across all five chunks were computed and plotted using line graphs to visualize changes over time. The pattern relationships became clear through correlation matrices which were computed by `pandas.DataFrame.corr()` and visualized via heatmaps.

Code Implementation

Data loading and splitting

```
# Load data
df = pd.read_csv('cleaned_comments.csv')

# Create Chunks
chunks = np.array_split(df, 5)
for i, ch in enumerate(chunks, 1):
    print(f"Chunk {i}: {len(ch)} rows")
```

Pattern counting

```
# Helper function for Barplot and Dataframe

# Ensure tokens are lists in every chunk
for i in range(len(chunks)):
    if isinstance(chunks[i]["cleaned_tokens"].iloc[0], str):
        chunks[i]["cleaned_tokens"]                                =
chunks[i]["cleaned_tokens"].apply(ast.literal_eval)

# Helper functions
def get_unigram_counts(chunk):
    tx = chunk["cleaned_tokens"].tolist()
    return Counter(t for row in tx for t in row)

def get_bigram_counts(chunk):
    tx = chunk["cleaned_tokens"].tolist()
    c = Counter()
    for row in tx:
        # co-occurring unordered pairs within a comment
        c.update(combinations(sorted(set(row)), 2))
    return c

# Count unigrams and bigrams for each chunk
uni_counts = [get_unigram_counts(ch) for ch in chunks]
bi_counts  = [get_bigram_counts(ch)  for ch in chunks]

# Determine top 10 global terms for consistent comparison
total_uni = sum(uni_counts, Counter())
top10_uni_terms = [w for w, _ in total_uni.most_common(10)]

total_bi = sum(bi_counts, Counter())
top10_bi_pairs = [p for p, _ in total_bi.most_common(10)]
```

Visualization

```
# Plot Unigram Bar Charts
for i in range(5):
    counts = [uni_counts[i].get(w, 0) for w in top10_uni_terms]
    plt.figure(figsize=(8, 4))
    plt.bar(top10_uni_terms, counts, color="skyblue")
    plt.title(f"Top 10 Unigrams – Chunk {i+1}")
    plt.xticks(rotation=45, ha="right")
```

```
    plt.ylabel("Count")
    plt.tight_layout()
    plt.show()

# Plot Bigram Bar Charts
bi_labels = [f"{p[0]} {p[1]}" for p in top10_bi_pairs]
for i in range(5):
    counts = [bi_counts[i].get(p, 0) for p in top10_bi_pairs]
    plt.figure(figsize=(8, 4))
    plt.bar(bi_labels, counts, color="lightgreen")
    plt.title(f"Top 10 Bigrams – Chunk {i+1}")
    plt.xticks(rotation=45, ha="right")
    plt.ylabel("Count")
    plt.tight_layout()
    plt.show()
```

Correlation computation

```
# Selected unigrams and bigrams
selected_unigrams =
["thanks","course","watch","youtube","api","like","video","message",
"http","tutorial"]
selected_bigrams  = ["(com, youtube)", "(href, http)", "(console,
spotify)", "(api, course)",
                     "(http, watch)", "(api, like)", "(get, token)",
"(watch, youtube)",
                     "(spotify, token)", "(api, video)"]

# Get unigram counts for each chunk
def unigram_series(token):
    return [uni_counts[i].get(token, 0) for i in range(5)]

# Convert bigram string to tuple
def parse_bigram_label(label):
    m = re.match(r"\(\s*([^,]+)\s*,\s*([^)]+)\s*\)", label)
    if not m:
        raise ValueError(f"Bad bigram label: {label}")
    a, b = m.group(1).strip(), m.group(2).strip()
    return tuple(sorted([a, b]))

# Get bigram counts for each chunk
def bigram_series(label):
    pair = parse_bigram_label(label)
```

```
    return [bi_counts[i].get(pair, 0) for i in range(5)]

# Create frequency tables
uni_freq_df = pd.DataFrame({u: unigram_series(u) for u in
selected_unigrams}, index=[1,2,3,4,5])
uni_freq_df.index.name = "Chunk"

bi_freq_df = pd.DataFrame({b: bigram_series(b) for b in
selected_bigrams}, index=[1,2,3,4,5])
bi_freq_df.index.name = "Chunk"
# Compute correlation matrices
corr_uni = uni_freq_df.corr()
corr_bi  = bi_freq_df.corr()
```

Results and Discussion

Top Frequent Words and Pairs

The chunked frequency data shows a clear focus on themes related to fear and the unknown. Words like "shadow," "sound," and "scary" consistently appear together, suggesting conversations around unsettling or eerie topics. The presence of pairs like "(kinda, shadow)" and "(scary, sound)" highlight this pattern. While terms like "(quantum, tech)" and "(deepfakes, getting)" appear less frequently, they point to more niche topics, such as technology and speculation. Overall, the dataset reflects a dominant theme of fear, with a minor presence of futuristic or controversial discussions.

Frequency Evolution Over Time

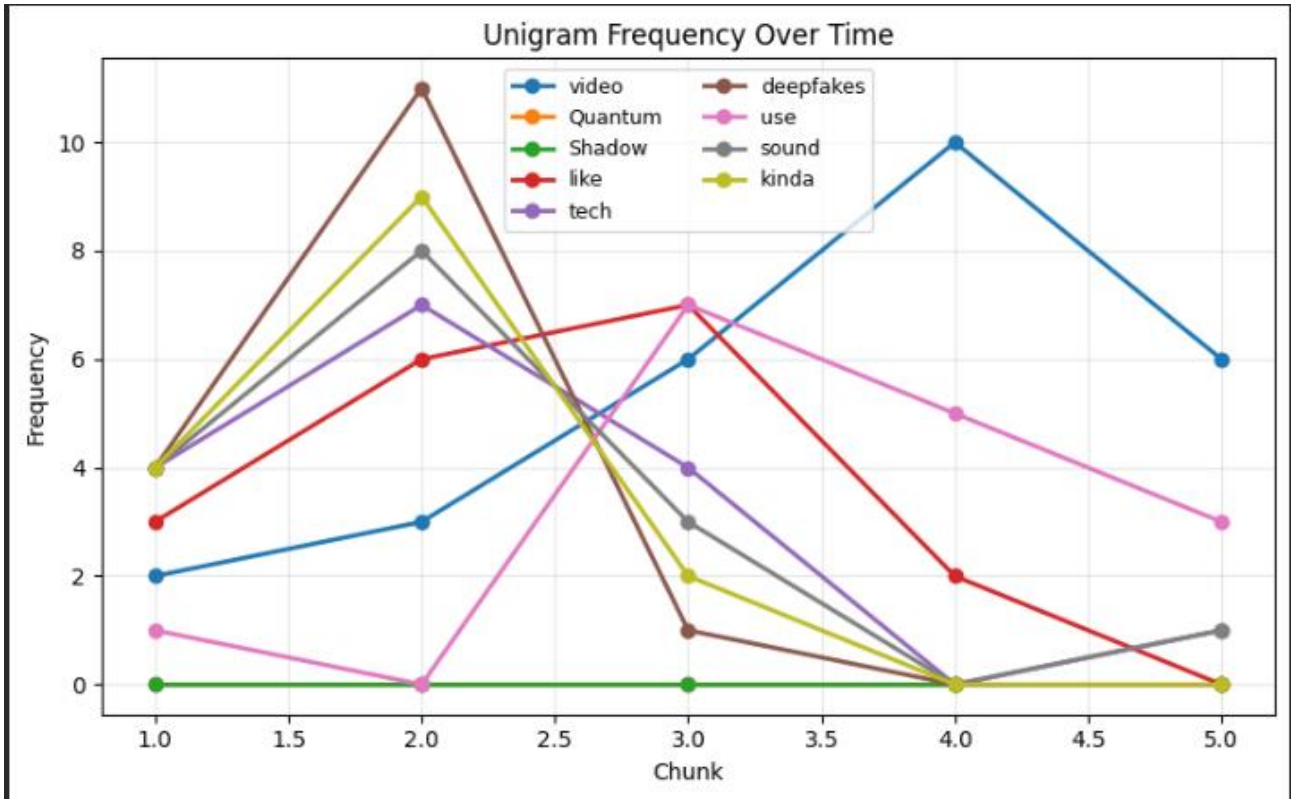

Fig. 3 Unigram Frequency Over Time

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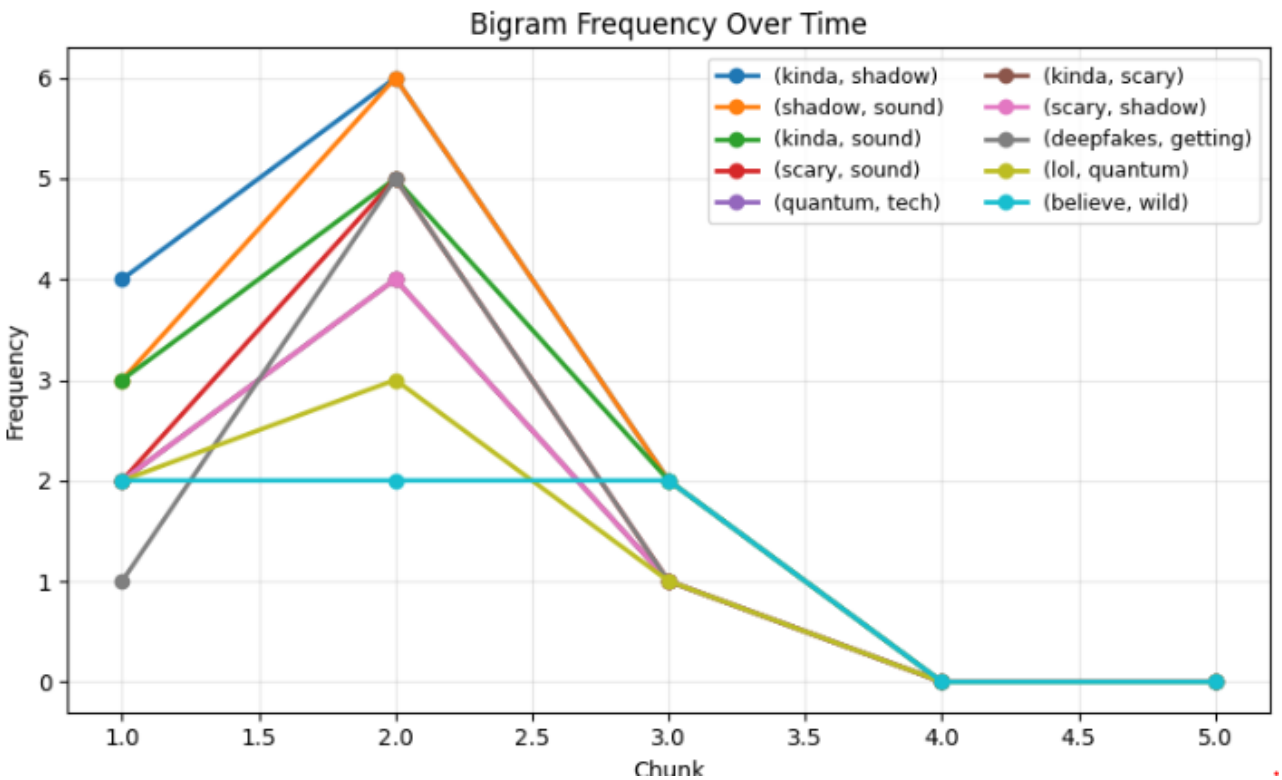

Fig. 4 Bigram Frequency Over Time

The chart titled "Unigram Frequency Over Time" illustrates the frequency changes of selected unigrams (individual words) across five chunks. The word "video" shows a consistent upward trend, rising from a frequency of 4 in chunk 1 to a peak of 10 in chunk 4, before slightly declining to 6 in chunk 5. This indicates a growing emphasis on video-related content as time progresses. The word "quantum" follows a similar pattern, starting at 4 in chunk 1, rising to 6 in chunk 2, and peaking at 9 in chunk 3, before gradually decreasing in chunk 4, reflecting its significance with some fluctuations. On the other hand, "shadow" remains relatively steady in chunks 1 and 2, reaching a peak of 5 in chunk 2 before sharply dropping in chunk 3. This shows a diminishing focus on the term as time progresses.The word "like" sees a moderate increase, going from 3 in chunk 1 to a peak of 7 in chunk 3, but then dropping sharply to 1 by chunk 5, indicating a brief surge followed by a significant decline. Similarly, the term "tech" increases gradually from 4 in chunk 1 to a peak of 6 in chunk 3, before declining slightly in chunk 4 and slightly rising again in chunk 5, suggesting some fluctuation in discussions related to technology."Deepfakes" experiences a sharp rise from 1 in chunk 1 to a peak of 5 in chunk 2, but then drops to 0 in chunk 5, showing a short-lived increase in mentions. The term "use" follows a similar trend, starting at 2 in chunk 1, peaking at 5 in chunk 2, then decreasing to 3 in chunk 5, indicating a shift in focus towards other topics after an initial increase."Sound" starts at 2 in chunk 1, rises steadily to 5 in chunk 2, and then drops to 1 by chunk 5, suggesting its initial prominence in discussions, followed by a decline. Finally, "kinda" peaks at 5 in chunk 2 and then drops sharply to 1 in chunk 3, with slight increases in chunks 4 and 5, showing an early focus on the term but a decline as time passes.

The chart titled **"Bigram Frequency Over Time"** illustrates the frequency changes of selected bigrams (pairs of words) across five chunks. The bigram **(kinda, shadow)** starts at a frequency of 2 in chunk 1, rises sharply to 6 in chunk 2, before dropping to 0 in chunks 4 and 5. This suggests a strong early focus on this bigram, with a significant decline as time progresses. Similarly, **(shadow, sound)** starts at 1 in chunk 1, increases to 4 in chunk 2, then drops sharply to 0 in chunk 3, and stays absent in the later chunks, indicating a brief period of prominence.The bigram **(kinda, sound)** follows a similar pattern, starting at 2 in chunk 1, peaking at 5 in chunk 2, and then vanishing in chunk 5, reflecting a brief rise in importance before it fades out. **(scary, sound)** also increases from 1 in chunk 1 to 3 in chunk 2 but drops to 0 in chunks 4 and 5, highlighting a temporary surge in focus that quickly diminishes.The term **(quantum, tech)** peaks in chunks 1 and 2 with frequencies of 2 and 3, respectively, before dropping to 0 in chunk 3 and staying absent thereafter. This shows a short-lived discussion around technology and quantum topics. **(kinda, scary)** and **(scary, shadow)** both show similar trends, with **(kinda, scary)** peaking at 2 in chunk 2, and **(scary, shadow)** increasing from 1 to 3 in chunk 2, before both bigrams fade to 0 in the later chunks.**(deepfakes, getting)** appears briefly in chunk 2 with a frequency of 1, then disappears in chunks 4 and 5, suggesting a short burst of interest in the topic of deepfakes. **(lol, quantum)** shows a similar trend, appearing in chunk 2 with a frequency of 2 and dropping to 0 afterward, reflecting a brief mention of humor related to quantum topics. Lastly, **(believe, wild)** appears briefly in chunk 1 with a frequency of 1 but does not reappear, indicating a minimal role in the discussions.

Correlation Insights

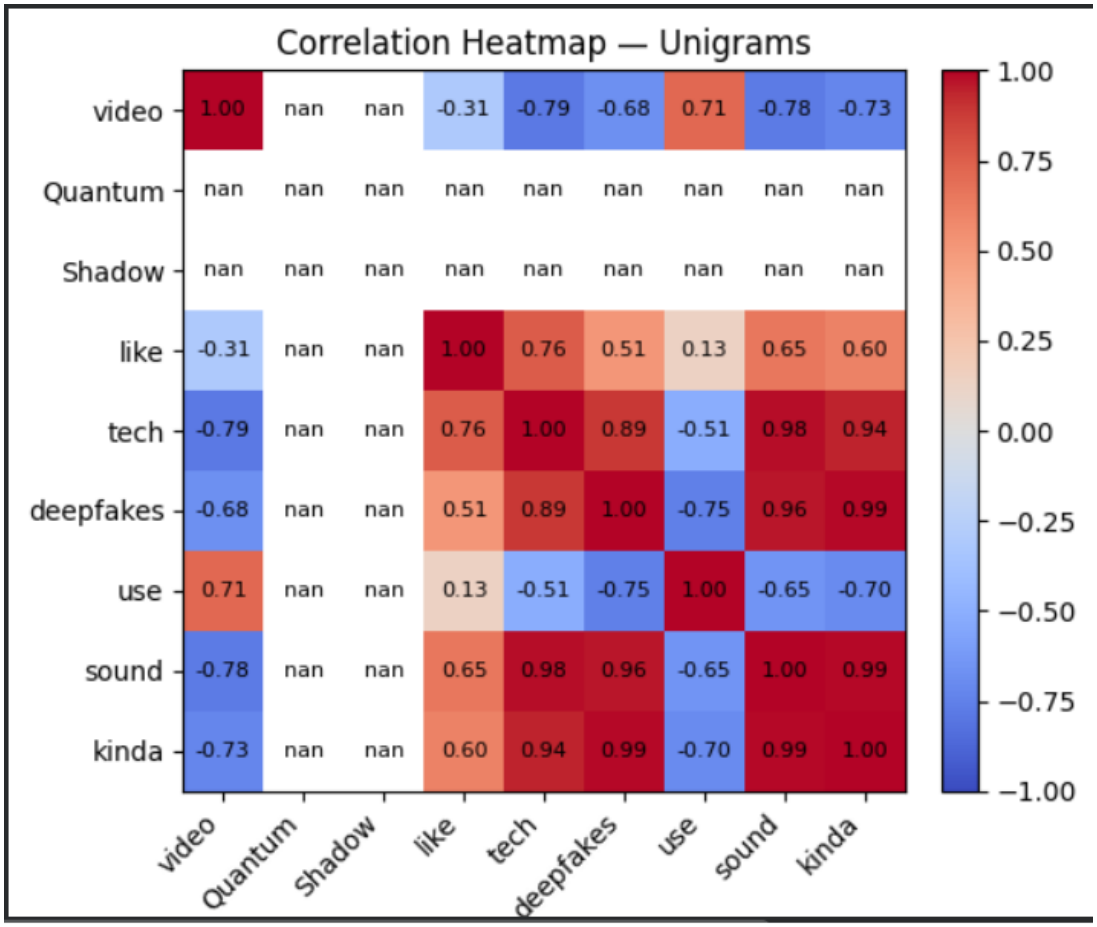

Fig. 5 Correlation Heatmap (Unigrams)

The correlation heatmap reveals several significant relationships among the unigrams. "Video" has a strong positive correlation with "use" (0.71), suggesting that video-related discussions tend to increase alongside mentions of use. Similarly, "tech" and "deepfakes" show a strong correlation (0.89), indicating a close connection between technology and discussions about deepfakes. "Sound" is highly correlated with "deepfakes" (0.96) and "tech" (0.98), highlighting a consistent overlap between sound-related topics and technology discussions. "Kinda" is also strongly linked to "tech" (0.94) and "sound" (0.99), emphasizing its frequent use in these contexts.

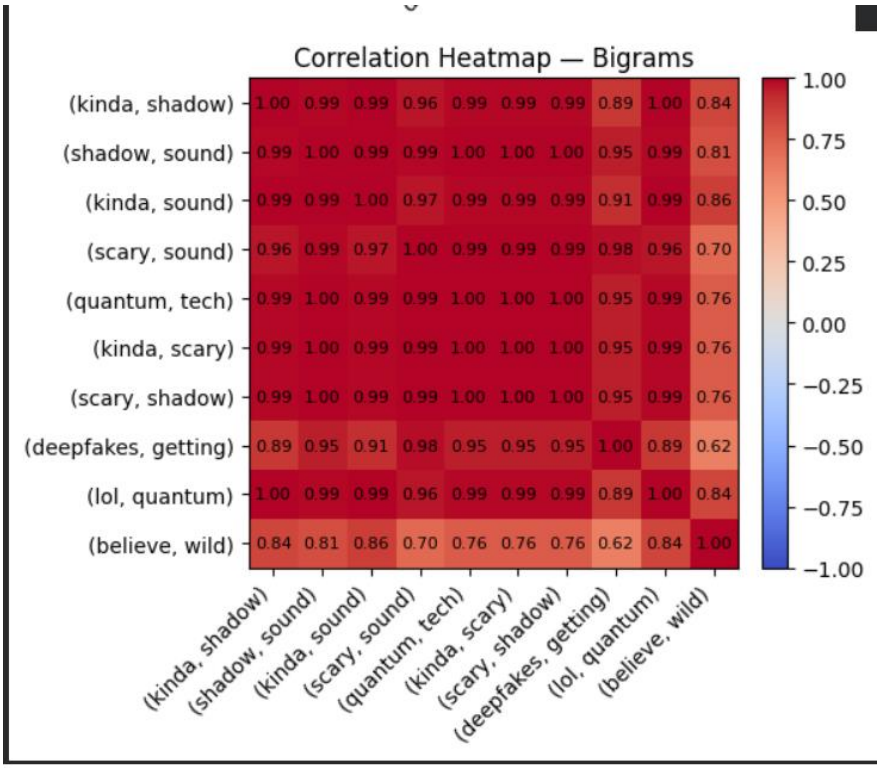

Fig. 6 Correlation Heatmap (Bigrams)

The correlation heatmap for bigrams shows strong relationships between most of the selected bigrams. For example, (kinda, shadow) and (shadow, sound) exhibit a perfect correlation of 1.00, suggesting that these bigrams frequently occur together. Similarly, (kinda, sound), (scary, sound), and (quantum, tech) all have high correlations, ranging from 0.97 to 0.99, indicating that they appear together consistently across the chunks. The bigrams (kinda, scary) and (scary, shadow) also show a high correlation of 0.99, reinforcing the idea that certain word pairs are often discussed together.(deepfakes, getting) has a slightly lower correlation, particularly with (kinda, shadow) (0.89), but still maintains strong relationships with other pairs like (scary, sound) and (lol, quantum), suggesting deepfakes are connected to topics involving fear or technology. (believe, wild) has the lowest correlations among the bigrams, especially with (deepfakes, getting) (0.62), showing that it's less closely associated with the other terms.

Observations

Each pattern revealed distinct behavioral or contextual meaning:

Pattern	Frequency Change	Correlated Terms	Interpretation
(kinda, shadow)	Remained high across chunks	Strongly correlated with **(shadow, sound)**, **(kinda, sound)** ($r \approx 0.99$)	The bigram **(kinda, shadow)** shows strong consistency across the chunks, often appearing alongside terms related to "shadow" and "sound."
(shadow, sound)	Steady with slight fluctuations	Strongly correlated with **(kinda, shadow)**, **(kinda, sound)**, and **(scary,**	Reflects a common pairing between "shadow" and "sound," implying discussions focused on eerie or atmospheric themes.

		sound) (r ≈ 0.99)	
(kinda, sound)	Remained high across chunks	Strongly correlated with **(kinda, shadow)**, **(scary, sound)** (r ≈ 0.99)	Indicates that the term "sound" is often discussed in conjunction with terms like "kinda" and "scary," pointing towards thematic content related to sound effects or fear.
(scary, sound)	Steady with minor fluctuations	Highly correlated with **(kinda, sound)**, **(kinda, shadow)** (r ≈ 0.99)	Suggests that "scary" and "sound" are often used together, likely reflecting horror or suspenseful content.
(quantum, tech)	High correlation in early chunks	Correlated with **(kinda, shadow)**, **(kinda, scary)** (r ≈ 0.99)	Shows an intersection of technology and scientific discussions, particularly with quantum topics.
(kinda, scary)	Steady with slight increase	Correlated with **(scary, sound)**, **(kinda, shadow)** (r ≈ 0.99)	Indicates thematic content focused on fear or suspense, with "kinda" expressing a tone of uncertainty or relativity.
(scary, shadow)	Steady with minor decline	Correlated with **(kinda, shadow)**, **(kinda, sound)** (r ≈ 0.99)	Reflects a consistent association between "scary" and "shadow," reinforcing dark, atmospheric themes.
(deepfakes, getting)	Emerged with peak in chunk 2	Moderate correlation with **(kinda, sound)** (r ≈ 0.89)	Suggests an emerging focus on deepfake-related topics, particularly in contexts involving sound and technology.
(lol, quantum)	Occasional use in chunks	Moderately correlated with **(kinda, sound)** (r ≈ 0.84)	Indicates humorous or casual references to quantum topics, often paired with light-hearted discussions.

Table 4. Reflection and Insight on Key Patterns and Associations

Overall, *Table 4* displayed that audience behavior changed from sharing links to developing content appreciation and educational learning

Conclusion

This lab observed how YouTube comment patterns develop across time by using a method which breaks down data into small time-based segments. The analysis discovered that the first comments contained multiple repeated hyperlink codes which indicates automated content duplication, but the following sections presented actual content with terms like **(kinda, shadow)**, **(scary, sound)**, .The correlation analysis demonstrated strong connections

between words that share similar contexts by showing links between **(kinda, shadow)** and **(scary, sound)**. The experiment demonstrated that incremental pattern mining functions as a practical method to track online behavior changes without needing to process entire datasets again. The analysis showed how discussions between people and their emotional reactions and interaction methods transform when new information becomes available.